

B-IoMV: Blockchain-based onion routing protocol for D2D communication in an IoMV environment beyond 5G

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ABSTRACT

Blockchain technology's popularity in terms of security, privacy, traceability, and trust is being applied in various major applications concerning connected autonomous vehicles. One of the most sensitive applications is military operations. We do not compromise with either security, privacy, trust, or communication latency of connected military vehicles, i.e., Internet of military vehicles (IoMVs). Achieving anonymity along with the security and privacy of sender, receiver, and data path for IoMVs is still an open question. Efforts we have made to address the aforementioned issue by proposing a blockchain-based onion routing protocol for IoMVs, i.e., B-IoMV, to achieve secure, trusted, and anonymous D2D communication. We studied the working of onion routing and presented a blockchain and token-based solution to strengthen the security and anonymity of IoMVs. A blockchain-based solution is quite costly and we use InterPlanetary File System (IPFS) to make the proposed B-IoMV system cost-effective. Finally, results show that the proposed B-IoMV system achieved better communication latency, data storage cost, and network bandwidth utilization.

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1. Introduction

In recent years, road fatalities have been increased a lot due to traffic crashes. As per the world health organization (WHO), the approximate number of humans who lose their lives each year due to road crashes is 1.35 million [1]. Fig. 1 shows the number of road fatalities from 2015 to 2020 of countries India, US, and UK. These fatalities were either due to human error or vehicle impairment. One of the possible solutions to mitigate the aforementioned issue (i.e., minimizing road fatalities) is the introduction of an intelligent transportation system (ITS), i.e., Internet of things (IoT)-based smart and connected vehicles (SCV) [2,3]. SCV's are driverless vehicles having various applications in public as well as private sectors such as traffic management, military operations, carriage holders, message passing, and many more. The most focused application of SCV is military operations, which can be used for patrolling in the most sensitive border areas and exchange information with the military base station. For such purposes, multiple SCVs (also called the Internet of military vehicles (IoMV)) are required for cooperative data exchange as in device to device (D2D) communication [4].

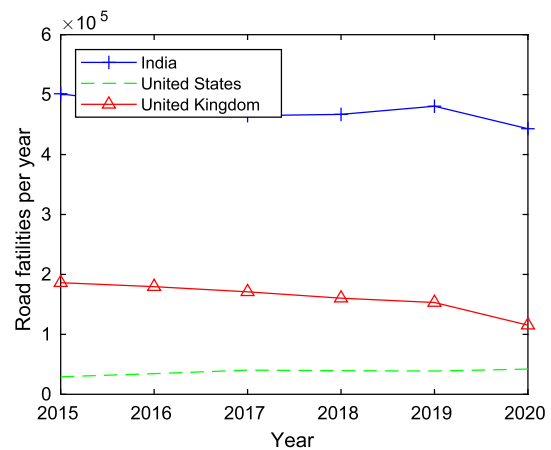


Fig. 1. Number of road fatalities from 2015 to 2020 of countries India, US, and UK [5–7].

IoMVs help save soldiers' lives as it can be controlled remotely by the base station over the wireless communication channel, which is open in nature and vulnerable to various security threats such as data sniffing, data modification, and denial of service.

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The aforementioned security threats can reveal the information related to the IoMVs, such as sender address, receiver address, secret data, and the path. It also causes packet delay and packet drop to the path from source to destination. The IoMVs carry the country's highly sensitive information and they should be capable enough to handle such attacks. The potential solution to deal with these security issues is integrating blockchain technology with the IoMV network. Blockchain offers data immutability (i.e., tamper-resistance) and traceability for the data packets moving from source IoMV to the destination IoMV/base station [8]. It also eliminates third-party systems' requirements to ensure the trust and security of sensitive data during its exchange over the network between IoMVs. It is an immutable database with history and no central point of attack [9]. Several other reasons that motivate us to integrate blockchain technology in the IoMV systems, which are (i) it helps to manage secure digital relationships between IoMVs, (ii) eliminate the third-party system (or middleman), which is quite time-consuming and cost-inefficient for data processing at IoMVs, (iii) transactions security among multiple participating IoMVs, (iv) offers transparency in data sharing among IoMVs, and (v) proper documentation. Here, the transactions in blockchain refer to the interaction between IoMVs and the military base station [10]. Despite many aforementioned blockchain benefits, anonymity is still questionable. Reid et al. [11] presented that the attacker can disclose the identity of any blockchain member by scrutinizing his/her public transactions.

Onion routing is a possible solution to achieve anonymity in the IoMV system for the nation's security. It forwards the message to multiple proxies that efface the relation between the data packet and the sender. In onion routing, a message is encased with many encryption layers (i.e., N) at the sender node and forwards to the next onion router/IoMV in the path [12]. Each onion router peels one case of encryption with its shared secret key and forwards the peeled onion (with $N-1$ encryption cases are left) to the next onion router/IoMV. This process is iterated until all encrypted cases are removed. The last encrypted case is received at the destination IoMV, which peels off the final onion layer and fetches the shared secret message [13]. This onion routing protocol maintains anonymity in the IoMV system, which helps exchange highly sensitive military information [14].

Given the aforementioned benefits of blockchain and onion routing protocol, in this paper, we propose a B-IoMV system, a blockchain-based onion routing protocol for D2D communication in an IoMV environment [15]. B-IoMV maintains the security of IoMV data along with the anonymity of sender and receiver IoMVs. The proposed system is ethereum-based, which is having high data (or transaction) storage cost. To minimize it, the proposed B-IoMV system uses a distributed InterPlanetary File System (IPFS), which is off-chain storage to store data in the blockchain environment. IPFS lead to higher throughput and low latency data access. It is a peer-to-peer system, which connects the network of the same system of files organized as a generalized Merkle-DAG (connected using hash values). The major benefits of IPFS are removal of duplicate values, removal of dependencies, reduces bandwidth necessity, archiving historical records, and faster data browsing [16,17]. The proposed system offers high security and anonymity, but the latency and reliability required between IoMVs for such highly sensitive data is not up to the mark. It is due to the traditional communication channel characteristics, i.e., 3G, 4G-LTE, LTE-A, and 5G.

As the IoMVs are smart, connected, and autonomous military vehicles, which require visible light communications, i.e., ultra-reliable low-latency communication between the IoMVs to make instant decisions such as when to press break, when to take a turn, instantly pass the enemy's information at the base camp, firing decisions, and many more. Traditional communication channels have high latency and low reliability, which does not meet

the requirements of IoMVs. 4G-LTE has a latency of ≈ 50 milliseconds, whereas LTE-A has < 10 milliseconds. Then, 5G has tried to minimize the network latency to < 5 milliseconds (end-to-end latency). The delay of even 1 millisecond in taking the military decisions by IoMVs is not acceptable. Such an issue can be solved with the inclusion of beyond 5G communication networks having massive ultra-low latency of < 1 milliseconds (end-to-end latency) and 10–100 μ sec (on-air latency) [18].

So, the inclusion of blockchain in onion routing with beyond 5G network in the proposed D2D-based IoMV system, i.e., B-IoMV, strengthen the communication in terms of security, anonymity, and ultra-reliable low-latency communication.

1.1. Motivation

Motivation of the paper is as follows.

- The existing SCV systems majorly focused on the security and privacy of the data exchange. A few researchers who considered the anonymity of the source, destination, and the data path along with their security and privacy issues. As per the literature, there is no approach to security, privacy, trust, and anonymity of SCVs in military operations. Thus, there is a need for such systems for military operations.
- SCVs exchanges data among other SCVs for making decisions. But, latency plays a vital role in such data exchanges if the application scenario is military operations. The current research works in SCVs for military applications have not been explored to the fullest. Hence, there is a need for a system that considers latency issues and the security, privacy, trust, and anonymity of data and path.
- Blockchain is a perfect candidate to meet the aforementioned requirements of security, privacy, trust, and anonymity of data exchanges in IoMVs. Thus, there is a need for a blockchain-based IoMV system to assure the above-mentioned parameters.

1.2. Research contributions

Following are the crisp contributions of the paper.

- A secure, trusted, and decentralized ethereum-based anonymous system, i.e., B-IoMVs for IoMVs, is proposed in beyond 5G environment.
- We present an encrypted verifying token-based solution to strengthen the onion-routing protocol security and anonymity.
- A secure and cost-effective data access procedure is presented using IPFS storage protocol.
- Performance evaluation of the proposed B-IoMV by considering communication latency, data storage cost, and network bandwidth utilization.

1.3. Paper organization

The rest of the paper is organized as follows. Section 2 discussed the related work and presented the relative comparison of the proposed system with the state-of-the-art systems. Section 3 presents the system model and problem formulation of the proposed system. Section 4 gives the proposed solution in terms of blockchain-based and storage-based solutions. Section 5 shows the performance evaluation of the proposed system. Finally, Section 6 gives the concluding remarks.

2. Related work

Many papers have been published to date by researchers across the globe on the security, privacy, and anonymity of connected ve-

Table 1

Comparison of existing approaches with the proposed one.

Author	Year	Objective	Blockchain?	Security framework?	Anonymity?	Latency issues?	Gaps
Hur et al. [19]	2017	Presented an onion routing-based solution to setup anonymous channels from source to destination	No	Yes	Yes	No	Network communication characteristics not focused
Biron et al. [2]	2018	Presented a real-time system to detect and estimate DoS attack in smart connected vehicles	No	Yes	No	No	Anonymity not preserved
Zhang et al. [8]	2019	Presented an onion routing-based solution to preserve the security and anonymity of blockchain-based applications	Yes	Yes	Yes	No	Not suitable for IoMVs in terms of network latency
Kuhn et al. [12]	2019	Presented the breaking of onion routing and given the solution for partially fixing it up	No	Yes	Yes	No	Not given a fully secure solution, which is not acceptable for IoMVs
Mancini et al. [20]	2020	Given a reference model to secure the autonomous vehicles used for military operations	No	Yes	No	No	Not presented the security and anonymity perspective
Haghighi et al. [21]	2020	Presented an onion routing-based highly anonymous system for vehicular ad-hoc networks	No	Yes	Yes	No	Not much focused on trust and latency issues in vehicular networks
Pei et al. [22]	2020	Presented an approach to traceback the IP addresses in an onion routing protocol using the system log files	No	No	Yes	No	It is time consuming and cost inefficient. Blockchain offers plausible solution
Buccafurri et al. [23]	2021	Presented an onion protocol to strengthen the anonymity of any system	No	Yes	Yes	No	Generalized explanation has given without targetting any application scenario
The proposed approach	2021	Presented a blockchain-based onion routing protocol for IoMVs in D2D mode underlying beyond 5G networks	Yes	Yes	Yes	Yes	-

hicular systems [19,2,8,12,20,21]. Most of them have focused on either security or privacy, but only a few researchers have given their thoughts on the anonymity of the sender, receiver, and data path. As per the literature, very few papers use blockchain to maintain the anonymity of SCVs in military operations. Hur et al. [19] Presented an onion routing-based solution to setup anonymous channels from source to destination with security. But, they didn't focus on the trust and network latency, and reliability issues. Then, Zhang et al. [8] presented a system called onionchain, which is a blockchain-based onion routing to preserve the security and anonymity of various applications. They have given generalized blockchain and onion routing-based solution for various applications, not solely focused on SCVs. Later, Haghighi et al. [21] presented an onion routing-based highly anonymous system for vehicular ad-hoc networks. Then, Pei et al. [22] given a solution to traceback the IP addresses in an onion routing protocol using the system log files. Then, the authors in [23] presented an onion routing protocol to strengthen the system's anonymity. Table 1 shows the comparative analysis of various existing approaches with the proposed survey.

As observed from the literature, the traditional systems have not used IPFS protocol to improve their efficiency. IPFS has various benefits such as, (i) it provides high throughput, low latency, data distribution, (ii) it is a peer-to-peer system, which connects the network of same system of files organized as a generalized Merkle-DAG, (iii) it removes duplicate data values, (iv) removal of dependencies, (v) reduces the bandwidth requirements, archiving historical records, and faster data browsing. The aforementioned benefits of IPFS improve the QoS of the proposed B-IoMV system.

3. System model and problem statement

3.1. System model

Fig. 2 shows the secret data exchange scenario consisting of multiple IoMVs, i.e., $\mathcal{V} = \{V_1, \dots, V_r, \dots, V_n\}$ and their connected V2V (or D2D) links $\mathcal{L} = \{L_1, \dots, L_i, \dots, L_l\}$, where $l < n$. V_1 wants to share the secret data (in the form of onion φ) with V_n , which is H hops away. An onion routing protocol is used to preserve the anonymity of the sender (against the eavesdropper and intermediate nodes) when the data is exchanged over multiple hops H towards destination node V_n . In this type of routing, the intermediate nodes are referred as onion router (O_R), which has an information only about the next (O_R^{next}) and previous (O_R^{prev}) hop onion routers O_R 's. In our scenario, the connected intermediate IoMVs are the O_R 's, which does not disclose the secret information to the eavesdropper except its O_R^{next} and O_R^{prev} . In this, the sender IoMV is a client node that generates the secret keys $\mathbb{K} = \{K_{V_1}, \dots, K_{V_r}, \dots, K_{V_n}\}$ for all the intermediate O_R 's using the Rivest-Shamir-Adleman (RSA) and Elgamal algorithms. Once the keys are generated, they are being shared with intermediate IoMV nodes/devices, i.e., O_R 's using the Diffie-Hellman key exchange algorithm. To enhance the data security and trust in anonymity (hiding sender's IoMV identity), a public blockchain network (β) with distributed, free, and immutable IPFS storage is used. A transaction is stored into the β if it satisfies the conditions of the smart contract (§). A transaction in blockchain is the ciphertext of verifying (VF) token and the secret data. The purpose of VF is to cross-verify the correct recipient (O_R^h) of the secret key K_h . The IPFS is used to save the data storage cost as well as data access latency. The

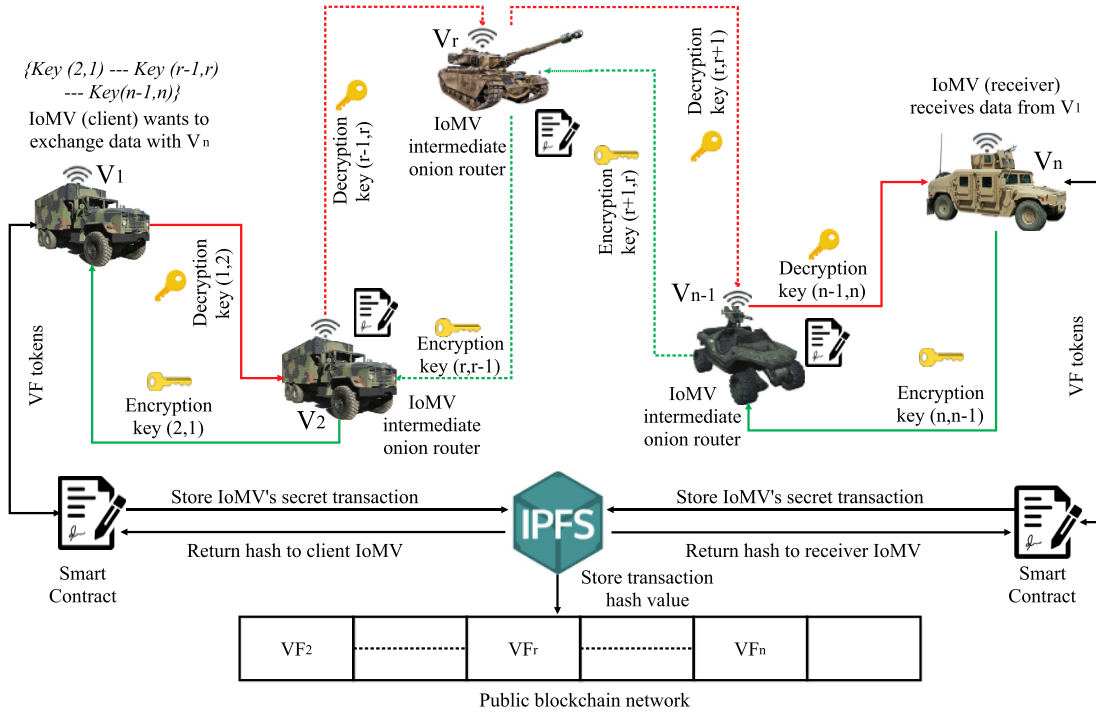


Fig. 2. B-IoMV system model.

proposed system model adds two-way security to the onion routing protocol to make it more secure and anonymous transmissions among IoMVs.

3.2. Problem formulation

In this subsection, the problem is formulated by considering the scenario mentioned in Fig. 2. The prime focus of this system is data security and sender and receiver anonymity. Before initiating the data exchange, the IoMV must have a sufficient energy (as the IoMVs are battery constraint vehicles) for the same. Otherwise, IoMV not able to participate in secret data exchange. Suppose M is the number of secret data packets to be exchanged between IoMVs, i.e., V_r and $\{V_1, V_2 \dots V_n\}$ under V2V/D2D communication such that $r \notin \{1, 2, \dots, n\}$. $\hat{h}_r$ is the estimation of left over energy of r^{th} IoMV V_r . Ψ is the threshold, which indicates the minimum energy required by any IoMV before the data exchange begins [24]. The estimated left over energy $\hat{h}$ is calculated as follows.

$$\hat{h}_r = \omega_T^{V_r} - \gamma_{(V_r, V_x)} \quad (1)$$

s.t.

$$\gamma_{(V_r, V_x)} = |M| \sum_{r=1}^n \sum_{x=1}^n \mathbb{E}_{(V_r, V_x)},$$

$$\{r, x\} \leq n,$$

$$r \neq x,$$

$$M \geq 0,$$

where ω_T is the total energy of IoMV and $\gamma_{(V_r, V_x)}$ is the total energy required for data exchange between IoMV V_r and V_x over M data packets. $\mathbb{E}_{(V_r, V_x)}$ is the energy required for data exchange over single data packet. If $\hat{h}_r \geq \Psi$ then IoMV can participate in data exchange else not.

Once it is verified that the IoMV has good amount of energy for data exchange, then the IoMV generates H 1024 bit secret keys ($\mathbb{K}$)

using the RSA algorithm [25]. The generated keys are represented as a set, which is shown as follows.

$$\mathbb{K} = \left\{ \frac{1 + d_1(\delta_1 - 1)(\Delta_1 - 1)}{\wp_1}, \frac{1 + d_2(\delta_2 - 1)(\Delta_2 - 1)}{\wp_2}, \dots, \frac{1 + d_h(\delta_h - 1)(\Delta_h - 1)}{\wp_h}, \dots, \frac{1 + d_H(\delta_H - 1)(\Delta_H - 1)}{\wp_H} \right\} \quad (2)$$

where $\{p_1, p_2, \dots, p_H\}$ and $\{q_1, q_2, \dots, q_H\}$ are random numbers, which are prime in nature. $\{d_1, d_2, \dots, d_H\}$ are the integer constants. $\{\wp_1, \wp_2, \dots, \wp_H\}$ are the public keys of intermediate onion routers O_R . The generated secret keys need to be shared with all onion routers O_R 's by the client IoMV for anonymous data exchange.

The secret keys $\mathbb{K}$ are exchanged with O_R 's using the standard Diffie-Hellman key exchange algorithm, which is an asymmetric key algorithm. It uses global prime element ζ and its generator (or primitive root) η to generate public and private keys to perform secure key exchange. Fig. 3 shows the dialogs for secret key exchange between client IoMV V_r and the intermediate onion router O_R 's. In onion routing, the current O_R^{curr} receives a request in the form of an onion ϕ from its previous onion router O_R^{prev} . It then decrypts the top most layer of onion ϕ with the shared secret key $\mathbb{K} \in \{K_{V_1}, \dots, K_{V_r}, \dots, K_{V_n}\}$ and get the identity (ID) of the next onion router O_R^{next} in path. It then passes the onion $\phi_{(layer-1)}$ to the O_R^{next} after padding by making length of $|O_R^{curr}|$ and $|O_R^{next}|$, i.e., $|O_R^{curr}| = |O_R^{next}|$. At last, the destination onion router O_R^{dest} receives the onion ϕ with only single layer, i.e., a layer after which the data is decrypted [13]. The destination onion router O_R^{dest} decrypts the original message ($\mathbb{M}$) and follows the same process of onion routing towards the requested router. Fig. 4 shows the visualization of key and data access by the onion routers in IoMV environment. Following equation depicts the key $\mathbb{K}$, ID, and onion to the next onion router O_R^{next} as data and so on till data packet or request reaches to the destination router O_R^{dest} .

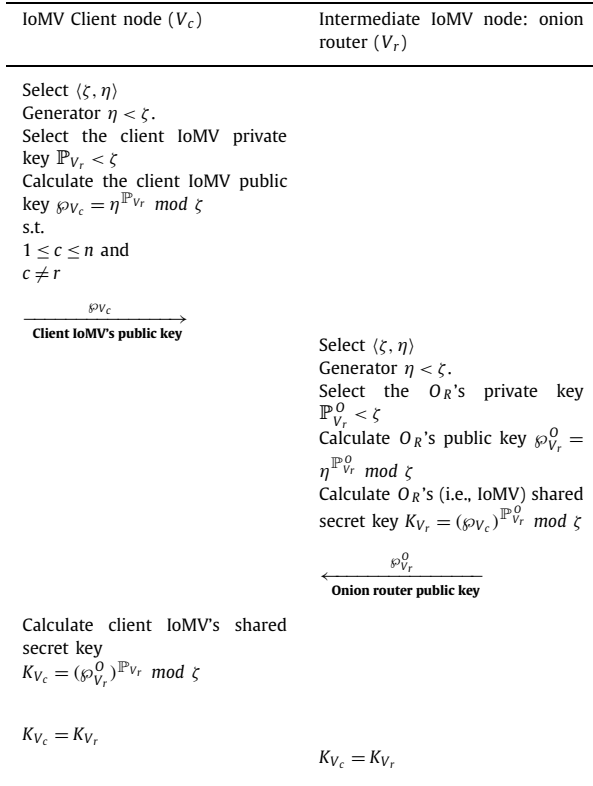


Fig. 3. Key exchange phase.

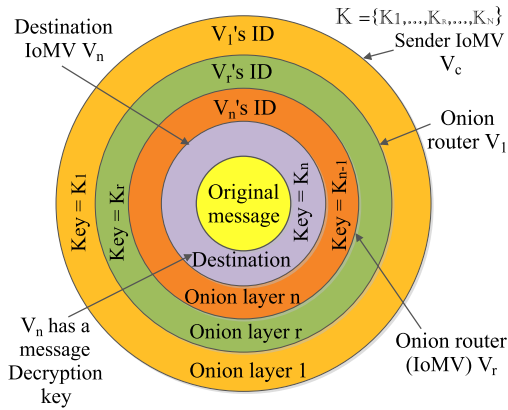


Fig. 4. Key and data access scenario of IoMVs in onion routing

$$\begin{aligned}
V_1(\underbrace{\mathbb{K}, O_R^{V_2}, \mathbb{M}}_{\text{Sender IoMV}}, \varphi) &\rightarrow \underbrace{O_R^{V_2}(O_R^{V_1}, K_{V_2}^\varphi, O_R^{V_3}, \varphi_{(Layer-1)})}_{\text{Second IoMV in route}} \dots \\
\rightarrow \dots &\underbrace{O_R^{V_r}(O_R^{V_{(r-1)}}, K_{V_r}, O_R^{V_{(r+1)}}, \varphi_{(Layer-(r-1))})}_{\text{rth IoMV in route}} \dots \rightarrow \\
\dots O_R^{V_n}(O_R^{V_{n-1}}, K_{V_n}^\varphi, \mathbb{M}) &\quad (3) \\
\underbrace{\hspace{10em}}_{\text{Destination IoMV}}
\end{aligned}$$

Onion routing in IoMV environment for secret data exchange ensures the anonymity of route (to whom the data packet is forwarding) by encrypting each layer of onion φ with the shared secret key of the respective onion router O_R , i.e., IoMV. Suppose any malicious IoMV takes over the genuine IoMV. In that case, the chances of a route from source to destination can be compromised, raising

Algorithm 1 Generation, conversion, and storage of VF token.

Input: Verifying token T_{VF} , shared secret key $\mathbb{K}$
Outputs: Encrypted VF token

Output: Encrypted VF token.

Initialization: Set the value of r as 1, i.e., $r=1$

1: **function** UNION_LAYER_ENCRYPTION($T_{VF}, \mathbb{K}$)

```

2: while (N(loMVs) > 0) do

```

```

3:    $T_{VF}^r(\text{Hex}) = \text{Random}()$ 
4:    $\text{Hex} = \text{Hex} \cup \{T_{VF}^r(\text{Hex})\}$ 

```

4: $T_{VF}^r = (\text{int})(T_{VF}^r(\text{Hex}))$
5: $T_{VF}^r = \text{Push_Key}(T_{VF}^r)$

```

5:    $T_{VF_{encrypted}}^r = \text{PubKey} \cdot T_{VF}^r$ 
6:    $\text{Store\_}T_{VF}^r \rightarrow \beta$ 

```

```

7:      r++
8:  end while

```

9: **end function**

the security and privacy concerns for highly confidential military data. In view of this, the problem statement or objective function of the proposed scheme is as follows.

$$\mathbb{O} = \max \left\{ \sum_{i=1}^M \sum_{j=1}^n (\text{security}(D_i), \tilde{h}_j) \right\} \quad (4)$$

s.t.

$$\Psi \leq \hbar_j \leq \omega_T, \forall 0 \leq j \leq n \quad (5)$$

where $security(D_{ij})$ represents the security of each data packet exchanged over the network and h_{ij} is the minimum energy required by any IoMV to transfer data packet to the destination IoMV.

4. Proposed solution

This section highlights the proposed solution to achieve the security of data packets mentioned in Eq. (4).

4.1. Blockchain-based solution

In this paper, we proposed a blockchain-based solution, which achieves the trust and security among the participating IoMVs. Our approach introduces the VF token (T_{VF}), which is encrypted and stored in the blockchain network β . The number of T_{VF} 's are less than or equals to the number of IoMVs/onion routers in the path from source IoMV to the destination IoMV, i.e., $|T_{VF}| \leq n$ and $T_{VF} = \{T_{VF}^1, T_{VF}^2, \dots, T_{VF}^r, \dots, T_{VF}^n\}$. The value of T_{VF}^r , $1 \leq r \leq n$ changes after each exchange or after specific ideal period of time. The token T_{VF}^r is randomly generated by the sender IoMV, which is an hexadecimal number. It then converted to the respective decimal number, i.e., integer and then used to encrypt the message or a layer of onion φ along with the respective layer's shared secret key $\mathbb{K}$.

Algorithm 1 shows the process of generation, conversion, and storage of VF tokens, where $N(IoMV_s)$ signifies the number of IoMVs in the scenario. T_{VF}^r (Hex) is the value of r^{th} VF token, which is hexadecimal in representation. $T_{VF_{encrypted}}^r$ is the encrypted r^{th} VF token that is encrypted using public key. Finally, $T_{VF_{encrypted}}^r$ stored into the blockchain. The complexity of Algorithm 1 is evaluated as $O(n)$, where n is the number of IoMVs.

Blockchain technology β is the best possible solution to securely share the encrypted token to the intermediate IoMVs or onion routers by the sender IoMV. It ensures trust, transparency, and protection against any kind of data modification. Blockchain is basically a chain of blocks that stores data as a transaction [26]. The blocks are accessed via a unique hash key $\mathbb{H}$, which are calculated using the hash value of the previous block, i.e., $\mathbb{H}_p$ as well as the current block data $\mathbb{B}_{data}^{curr}$ [27]. Hash is a one-way function that is represented as [28].

$$\mathbb{H} : \{\xi\} \rightarrow \{0, 1\}^k, \forall \xi \in M_{V_x, V_y} \quad (6)$$

$$\mathbb{H}(\xi) = \xi' \quad (7)$$

s.t.

$$\{V_x, V_y\} \in \mathcal{V} \quad (8)$$

$$\mathbb{H}(\xi') \neq \xi \quad (9)$$

$$\mathbb{H}(\xi_x) \neq \mathbb{H}(\xi_y), \xi_x \neq \xi_y \quad (10)$$

where ξ is a message to the transmitted that belongs to the set of messages M . ξ' is the hash value of any message ξ transmitted over wireless channel from V_x to V_y (between the onion routers). So, the r^{th} block hash is calculated as below.

$$\mathbb{H}_r = \{\mathbb{H}_{r-1} \oplus T_r \oplus Nonce_r\} \quad (11)$$

where $\mathbb{H}_{r-1}$ is the hash value of the previous block in the chain. T_r is the timestamp of the r^{th} block and $Nonce_r$ is the nonce value of the r^{th} block.

Once the token T_{VF}^r is generated for r^{th} IoMV and encrypted with the public key algorithm $T_{VF}^{r, encrypted}$, the proposed system is distributed to the r^{th} IoMVs/onion routers. Likewise, n number of tokens need to be generated for n IoMVs. Then the hash values $\mathbb{H}$ of the respective encrypted VF tokens are generated (as Eq. (6)) and stored as a transaction λ_r into the blockchain β . The VF token T_{VF}^r is used to decrypt the onion layer by the onion router or IoMV. The r^{th} encrypted VF token $T_{VF}^{r, encrypted}$ is to be decrypted with the private key of r^{th} onion router only. This ensures the privacy of data against the malicious nodes (onion router or IoMV). Following are the properties of the proposed VF token T_{VF} .

- It is randomly generated by the sender IoMV before initiating any communication with the destination IoMV.
- It is unique for each IoMV as they are having their own private keys.
- The VF token T_{VF} will expire after its first use by the receiver IoMV. Then, the sender generates another random token, encrypts it, and stores into the blockchain network β .

This one-time usage of VF token T_{VF} strengthens the security and privacy of data against the malicious IoMVs. The data decryption process at the intermediate and destination IoMVs are shown as below.

$$T_{VF}^{r, decrypted} = K_{V_r}^{private}(T_{VF}^{r, encrypted}) \quad (12)$$

$$\forall r \ 1 \leq r \leq n \quad (13)$$

where $K_{V_r}^{private}$ is the private key of r^{th} IoMV, which is used to decrypt the encrypted VF token. Once the VF token is decrypted $T_{VF}^{r, decrypted}$, it is then appended with the shared secret key to decrypt the message or onion layer. The modified shared secret key $\widehat{K}_{V_r}$ of any r^{th} IoMV is as follows.

$$\widehat{K}_{V_r} = K_{V_r} + T_{VF}^{r, decrypted} \quad (14)$$

Algorithm 2 shows the procedure to decrypt the layer of onion at the respective onion router or IoMV with the modified shared secret key $\widehat{K}_{V_r}$. Its computational complexity is $O(mn)$, where m and n are the number of IoMVs and encryption time respectively.

As $\mathcal{V} = \{V_1, \dots V_r \dots V_n\}$ or onion routers O_R in the proposed system are connected via a public blockchain network β , which need to be ensured that only valid IoMVs could join and access the network. To accomplish this, the proposed system needs a communication certificate (CC) from each IoMV who wants to access and

Algorithm 2 Procedure to decrypt the onion layer at IoMV.

Input: $T_{VF}, K_{V_r} \in \mathbb{K}, K_{V_r}^{private}$
Output: Decrypted data packet or onion layer

```

1: procedure BLOCKCHAIN_SECURITY( $T_{VF}, K_{V_r} \in \mathbb{K}, K_{V_r}^{private}$ )
2:   while (True) do
3:     if ( $N(\text{IoMVs}) > 0$ ) then
4:       Call Onion_Layer_Encryption( $T_{VF}^r, K_{V_r}$ )
5:        $T_{VF}^{r, decrypted} = K_{V_r}^{private}(T_{VF}^{r, encrypted})$ 
6:        $\widehat{K}_{V_r} = K_{V_r} + T_{VF}^{r, decrypted}$ 
7:       Decrypt_Data_Packets( $D_i, \widehat{K}_{V_r}$ )
8:     else
9:       Display ("No path identified")
10:    end if
11:  end while
12: end procedure

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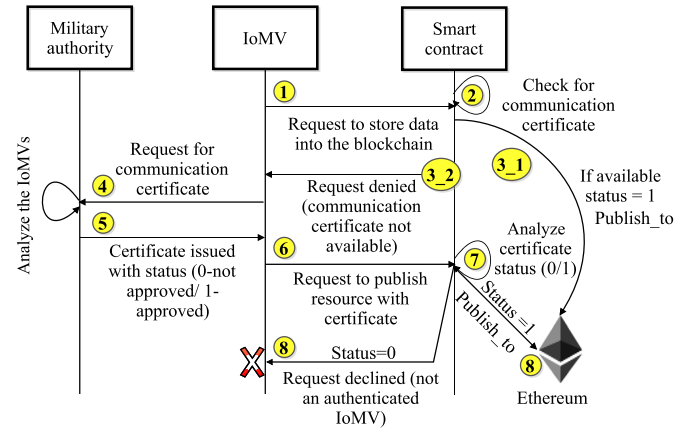


Fig. 5. Flow of IoMVs registration and publish phase.

store data into the blockchain network. Fig. 5 shows the procedure for IoMVs to get certificate from the authority is shown below.

The aforementioned registration process to get CC offers an additional security layer to the data/onion floating from source IoMV to the destination IoMV. This restricts the malicious IoMV to get register themselves with the blockchain network β . In any case, any malicious IoMV succeeded in getting the valid CC with status=1 from the registration authority by cloning the identity of valid IoMV. Then, the second security layer, i.e., decrypt the encrypted VF token $T_{VF}^{r, encrypted}$ with the valid IoMVs private key resist the malicious IoMV to access data from the blockchain β . This is possible because the sender IoMV knows the identities and public and shared secret keys $\mathbb{K}$ of all the onion routers O_R 's / IoMVs in the path from source to destination IoMVs.

The blockchain-based solution offers trust, security, privacy, and immutability. But, the cost of storing a transaction into the ethereum blockchain is quite high. The sequence of equations to calculate the cost to store one word, i.e., 256 bits in ethereum is mentioned as follows. In ethereum, the gas needed to write in ethereum's SSTORE is 20K Gas, i.e., $\mathbb{G} = 20K$ [29].

$$1 \text{ word} = 20000 \text{ Gas} \quad (15)$$

To store 1 KB of data, the gas required is

$$1 \text{ KB} = 2^5 \times 20000 \text{ Gas} \quad (16)$$

The existing price of gas ($\mathbb{G}_{pr}$) and ethereum ($\mathbb{Z}_{pr}$) are 185.4 gwei [30] and \$1672.52 (as on 27-Mar-2021) respectively. So, the cost to store l -words in ethereum is

$$1 \text{ Ethereum} = 10^9 \text{ gwei} \quad (17)$$

$$\text{Ethereum}_{l\text{-word}} = (l \times \mathbb{G}) / (10^9) \quad (18)$$

Algorithm 3 IPFS retrieval file procedure.**Input:** file location (θ), data hash value ($\mathbb{H}$), shared secret key K_{V_r} **Output:** Retrieval of r^{th} IoMV file $V_r \rightarrow D_r^{\text{file}}$

```

1: procedure IoMV_FETCH_FILE( $D_r^{\text{file}}$ ,  $\theta$ ,  $\mathbb{H}$ ,  $K_{V_r}$ )
2:   File_Location =  $\theta$ 
3:   Hash_Value =  $\mathbb{H}_{D_r^{\text{file}}}$ 
4:   if (Hash_Value  $\in D_r^{\text{file}}$  and  $K_{V_r}$ ) then
5:     if ( $\mathbb{H}_{\text{encrypted}} = \mathbb{H}_{D_r^{\text{file}}}$ ) then
6:       Return_Location_To_IoMV( $\theta$ )
7:     end if
8:   end if
9: end procedure

```

So, the cost to store l -words in USD is

$$\text{USD}_{l\text{-word}} = (\mathbb{G}_{pr} \times \text{Ethereum}_{l\text{-words}}) \times \mathbb{Z}_{pr} \quad (19)$$

4.2. Storage-based solution

To overcome the aforementioned issues of data storage cost in the blockchain network β , we incorporate an off-chain IPFS protocol for storing the data in B-IoMV. It is an immutable as well as distributed storage system with no single point of failure problem. It also helps to speed up the data downloading process by hosting part of the data to the other peers by utilizing minimum network bandwidth. The basic characteristics of IPFS protocol beneficial for B-IoMV are as follows.

- It is off-chain storage that does not put any burden on blockchain network β in terms of data volume.
- It is a complete distributed network and can serve as both server as well as a client.
- It offers high throughput as well as low latency in off-chain data access.
- It takes care of data duplicacy by generating the same hash value for the same data.
- It keeps data in the fixed-length blocks linked together as a generalized Merkle directed acyclic graph.

The above-mentioned characteristics of IPFS strengthen blockchain's suitability in onion routing protocol-based IoMV communication, i.e., D2D communication. Algorithm 3 shows the file retrieval procedure for IoMV from the IPFS storage [31]. The complexity of Algorithm 3 is evaluated as $O(n)$, where n is the number of IoMVs in the blockchain network.

5. Performance evaluation

This section presents the performance evaluation of B-IoMV considering performance metrics: communication latency, storage cost, system scalability, and IPFS bandwidth utilization. For blockchain implementation (Ethereum), the Solidity language has been used for writing the smart contracts and executed over the Remix IDE [32]. For publishing data onto the IPFS, we have used the IPFS desktop application with different peer IDs (corresponding to the data uploaded). The detailed discussion on these performance metrics is as follows.

5.1. Communication latency

Fig. 6 highlights the communication latency difference between the beyond 5G-assisted B-IoMV system and other traditional communication network infrastructures such as 4G, LTE-A, and 5G. The achievable latencies of 4G, LTE-A, and 5G networks are ≈ 50 milliseconds, < 10 milliseconds, and < 5 milliseconds, respectively. The beyond 5G network possesses a communication latency of

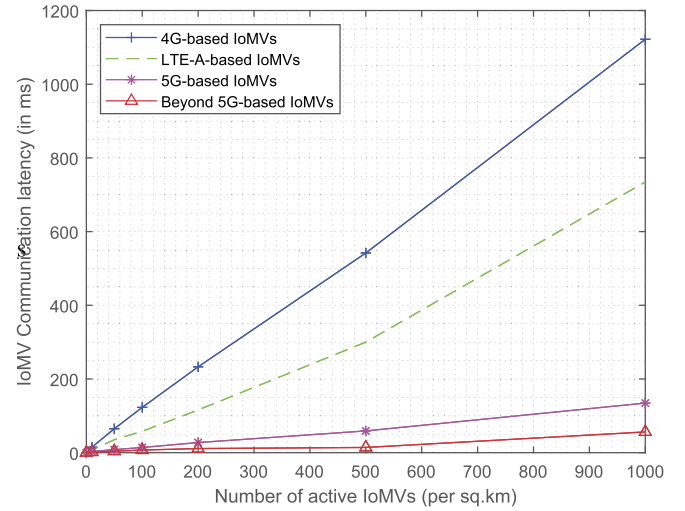


Fig. 6. Achievable communication latency of B-IoMV. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

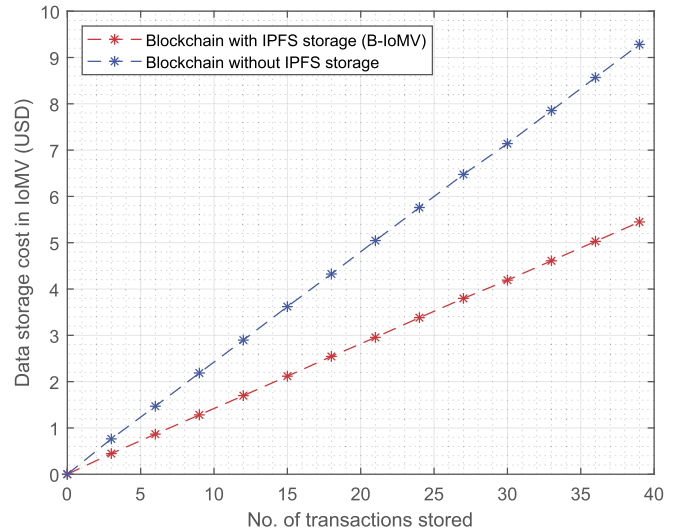


Fig. 7. Data storage cost of B-IoMV.

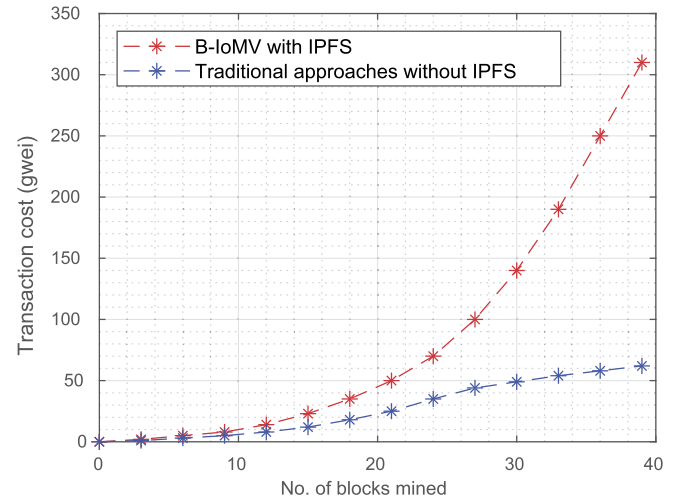


Fig. 8. B-IoMV scalability.

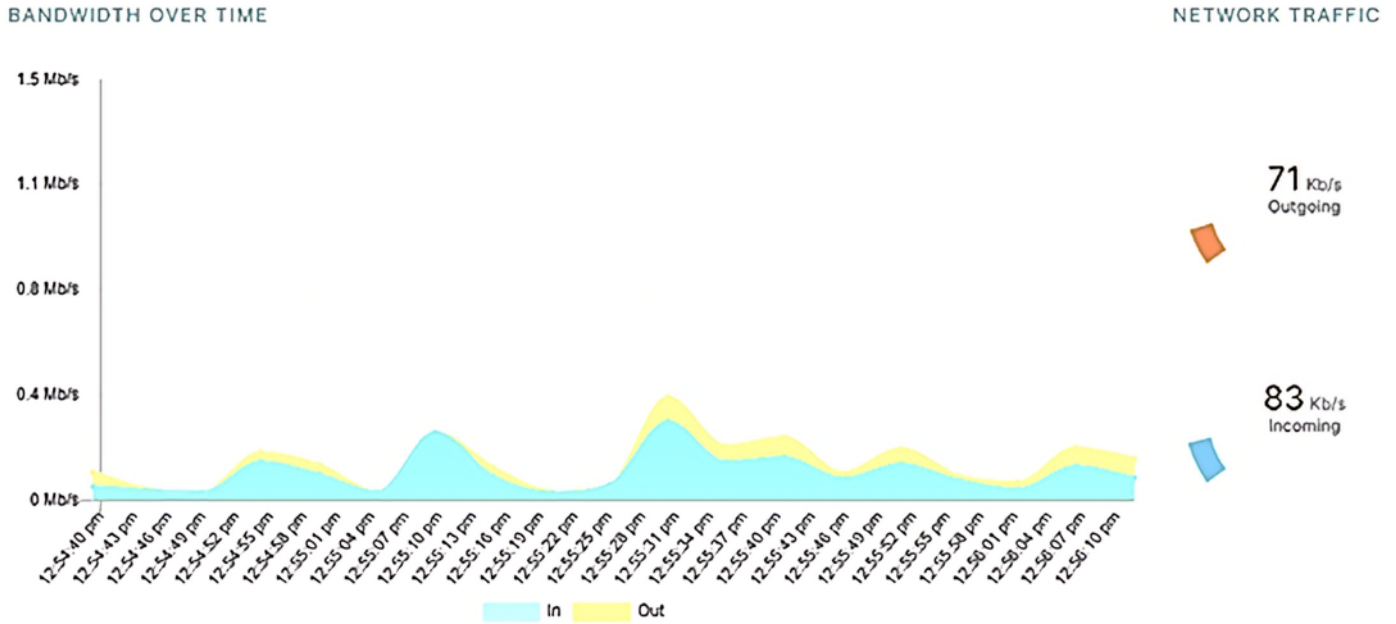


Fig. 9. Utilization of network bandwidth with IPFS.

< 1 millisecond, which is quite low and impressive in IoMVs communication. From Fig. 6, we can clearly see that the achievable communication latency of B-IoMV is quite low compared to the traditional network systems.

5.2. Data storage cost

Fig. 7 shows the data/transaction storage cost comparison of blockchain and IPFS-based IoMV systems. The blue line shows the storage cost of the non-IPFS ethereum-based IoMV system, which is quite high compared to the IPFS-enabled ethereum-based IoMV, as shown with the red line. The reason behind the differences in storage costs is mentioned in Section 4.

5.3. System scalability

Fig. 8 shows the scalability comparison of the proposed B-IoMV system with the traditional non-IPFS-based blockchain systems. IoMVs store their data into the IPFS protocol and the calculated hash of respective data is stored into the blockchain, whose size is quite less than that of the original data transaction size. Thus, B-IoMV offers more number of transactions to be appended to the blockchain network at the same time, which serves more number of IoMVs, hence improving the overall scalability.

5.4. IPFS bandwidth utilization

In B-IoMV, the IPFS protocol requires very low bandwidth as it is a distributed storage that breaks the original content into smaller chunks or blocks of fixed length. Fig. 9 shows the input and output network bandwidth utilization with IPFS simulations on local node/system. To perform this simulation, we first install and set up the IPFS node in the local system. After that, we upload and download the content by using the generated hash key. From Fig. 9, we can analyze that the IPFS helps to optimize the IoMV system.

6. Conclusion

This paper presents a blockchain-based onion routing protocol B-IoMV for the IoMV environment towards beyond 5G networks

considering security, privacy, anonymity, and trust. We give deep insights into the state-of-the-art blockchain or onion routing-based connected and smart autonomous vehicles. We exhibited the proposed system B-IoMV achieved ultra-low latency, minimized data storage cost, and network bandwidth utilization compared to the traditional systems. The performance evaluation shows the proposed B-IoMV incurs ultra-low latency compared to the traditional communication systems (i.e., 4G, LTE-A, and 5G) and also possesses lower data storage cost compared to conventional non-IPFS based blockchain systems by utilizing minimum network bandwidth.

In the future, we plan to deploy IoMV smart contracts in a real-world scenario with minimum latency and high reliability. We also incorporate intelligence at the receiver IoMV to filter out the malicious node's false requests as well as live streaming of video feeds for better insights.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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